MediFlow AI

An Intelligent Integrated Healthcare Ecosystem for Patient–Physician–Laboratory–Pharmacy Coordination

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1. Executive Summary

MediFlow AI is a web-based healthcare platform that connects **patients, doctors, laboratories and pharmacies** in one system. It replaces handwritten prescriptions, manual queues and paper lab reports with one digital flow: hospital and doctor search → appointment (or direct lab test) → triage → consultation → lab test → e-prescription → pharmacy → follow-up.

What makes it different from ordinary hospital software:

- **AI help for prescriptions.** A photo or typed draft is read by the AI, and local brand names (e.g. Napa, Seclo, Ciprocin) are matched to their generic names through the DGDA register.
- **Safety check before signing.** Every draft is checked against the patient's allergies, current medicines and vital signs.
- **Emergency-aware routing.** Critical patients go straight to the Emergency Room under a **Red Alert**; urgent patients get priority in OPD.
- **Direct lab tests.** For simple tests on an approved list, the patient can book the test directly, without seeing a doctor first.
- **Hospital and doctor directory.** Patients search hospitals, branches, departments and doctors, and see opening hours, emergency numbers and free slots.
- **Vitals with history.** Each vital-sign set is classified (normal / borderline / critical) with one common design, and saved so doctors see the trend.
- **85 % confidence rule, per medicine.** Items the AI is ≥ 85 % sure about are pre-filled; less certain fields are highlighted. A doctor always confirms and a pharmacist re-checks. The AI never signs or dispenses anything.

2. Problem, Objectives and Scope

Problem. In Bangladesh, healthcare coordination is mostly paper-based. Illegible prescriptions, manual queues, disconnected lab records and no emergency prioritisation cause medication errors, treatment delays and poor continuity of care. There is no safe, connected channel from hospital search and registration to dispensing and follow-up.

Objectives. (1) Digitise appointments and prescriptions end to end. (2) Connect patient, doctor, lab and pharmacy on one platform. (3) Improve medicine adherence with reminders. (4) Give all users real-time notifications. (5) Send critical emergencies to immediate care and urgent cases to priority OPD. (6) Reduce prescription errors with AI plus human review. (7) Let patients find a hospital or doctor and book simple lab tests directly. (8) Build a standards-based (HL7 FHIR R4) framework aligned with national systems (DGDA, DGHS, DHIS2).

Role	What they do in MediFlow AI
Patient	Searches hospitals and doctors, registers, books appointments and direct lab tests, sees queue token, views e-prescriptions and reports, receives reminders
Doctor (Physician)	Reviews history and vitals trend, consults, orders tests, writes and e-signs prescriptions with AI help
Laboratory	Receives doctor-ordered and direct test orders, tracks samples, uploads and verifies reports, flags abnormal results
Pharmacy	Validates and dispenses signed prescriptions, records dispensing
Nurse	Sets the triage level (Critical / Urgent / Normal), enters symptom severity and vitals, raises Red Alert, adjusts the urgent queue
Front desk	Registers walk-ins, books appointments and direct lab tests for patients, checks patients in, shows queue and token, flags a suspected emergency
Emergency team (ER)	Receives Red Alert with patient summary; sends back handover status
Administrator	Manages users, hospital and doctor directory, direct-test list and thresholds; reviews audit logs and analytics

Scope. In: hospital and doctor directory, outpatient coordination, emergency triage and routing, AI-assisted e-prescription, lab coordination (including direct tests), pharmacy dispensing, reminders. Out (this phase): inpatient beds, billing/insurance and online payment, radiology images, video consultation, autonomous diagnosis. The AI assists; it never makes final clinical decisions.

> **The idea in plain words:** photo → read → understand → check safety → doctor confirms → pharmacist re-checks. The AI only prepares a draft and highlights what needs attention.

3. Clinical Gap Analysis

Area	Today	Risk	MediFlow AI response
Prescription writing	Handwritten, illegible	Dispensing and dose errors	AI reads draft → structured e-prescription; doctor e-signs
Safety checking	Depends on memory	Missed allergies / interactions	Automatic check before signing
Vital signs	On paper, not trended	Deterioration missed	Auto classification + vitals history
Queues & emergencies	First-come; critical cases wait in the same line	Long waits, unsafe handling	Critical → Red Alert → ER; urgent → priority OPD
Hospital information	Phone calls, scattered websites	Wrong hospital or doctor, wasted trips	Hospital and doctor directory with slots and emergency numbers
Simple lab tests	Doctor visit needed first	Extra cost and waiting	Direct lab test booking from an approved list
Lab records	Paper, disconnected	Repeated tests, late diagnosis	Digital order, sample tracking, doctor or patient notified
Adherence	No reminders	Missed doses and visits	Smart reminders (SMS / email / push)
Data protection	Physical files, no access log	Privacy breach	Role-based access, encryption, audit trail

4. Existing Systems and Novelty

Capability	Paper practice	Standalone hospital software	Consumer booking apps	Global EHR suites	MediFlow AI
Patient–doctor–lab–pharmacy in one flow	No	Partial	Partial	Yes	Yes
Handwritten prescription digitisation (AI)	No	Rare	No	Rare	Yes
Draft contradiction check	No	Partial	No	Yes	Yes
Emergency routing (ER vs priority OPD)	No	Rare	No	Partial	Yes
Per-item confidence review	n/a	No	No	Varies	Yes (≥ 85 %)
Local brand-name (DGDA) mapping	No	Rare	No	No	Yes
Direct lab test booking by patient	No	Partial	Partial	Partial	Yes
Hospital and doctor directory with slots	No	Partial	Yes	Partial	Yes
FHIR interoperability	No	Varies	Rare	Yes	Yes (R4)
Low cost, suited to local clinics	Yes	Medium	Yes	No	Yes

The table compares general categories, not specific commercial products.

#	Novel contribution	Benefit
N1	AI pipeline for handwritten / typed prescriptions with DGDA brand → generic mapping	Removes illegibility; enables automatic safety checks
N2	Draft contradiction detection (allergy, interaction, dose, vitals) before signing	Catches errors at the cheapest point
N3	Vitals classification with one common design (polymorphism) and a stored history	Early warning; new vitals added without new code
N4	Per-item 85 % gate: pre-fill ≥ 85 %, highlight below, hard-stop on critical conflict	Doctors keep the useful output; AI cannot silently lower safety

#	Novel contribution	Benefit
N5	Emergency-aware queue: Critical → Red Alert → ER; Urgent → priority OPD	Faster care for critical patients
N6	Direct lab tests from an approved list, with abnormal-result advice to see a doctor	Faster, cheaper simple testing without losing safety

5. Hospital Website Review and New Features

To check that MediFlow AI fits real hospitals, seven public hospital websites were reviewed (home or guide pages, October 2026). The table lists what each site offers, a limitation visible without logging in, and the idea taken for MediFlow AI. A feature may exist behind a login; only what was visible is reported.

Hospital	Seen on the website	Limitation seen	Idea for MediFlow AI
Aurora Specialized Hospital, Dhaka	Online appointment page, 24x7 hotline, report page, health check-up packages, doctor search by body part	Departments, doctors and packages load dynamically; no single-test booking or queue status visible; reports on a separate page	Body-part / department search for new users; packages as test bundles
BIRDEM General Hospital, Dhaka	Find consultant, appointment form (department, doctor, phone, fee), hotline list (emergency, ICU, ambulance), online report	Department list mixes clinical and administrative units; online report opens a separate plain-HTTP address; some service cards show placeholder text	Clean directory of clinical departments only; one login for reports; hotlines kept on the hospital card
Evercare Hospital Dhaka	Find a doctor, request an appointment, online report portal, tele-consultation, health packages, visitor guide	Appointment is a “request”, not an instant slot; reports and tele-consult are separate portals; packages are fixed bundles; English only	Instant slot booking; patient picks single tests from the approved list; Bangla and English
Ibn Sina Trust (many branches)	Doctor search by department, branch or name; 25+ units; diagnostic services; check-up packages; patient-care counters	Report delivery and billing shown as counters (no online link on home page); doctor cards point to a hotline; branch and test availability not linked	Hospital / branch table; branch-wise doctor and test search
Square Hospitals, Dhaka	Online appointment via doctor list, online payment, 24-hour emergency and ambulance, satellite centres	Online payment on a separate plain-HTTP address (port 8080); one hotline for appointments, emergency and billing; no online report or direct test booking seen	Separate emergency path (Red Alert); online payment kept for a later phase
Singapore General Hospital	Appointments, video consult, order medicine online, personal health records (lab results, prescriptions), pay online, “I am a patient / visitor…” entry	Depends on national systems (HealthHub, MediSave) not available locally; many separate apps; very large menu	Role-based simple entry; patient view of own lab results and prescriptions; video consult and medicine delivery as future work
Mayo Clinic (Minnesota guide)	Request appointment, find a doctor, patient and visitor guide, interpreter services, language choice, mobile app, cost estimator, records request	Appointment is a request; billing and insurance model is US-specific; page is mostly visitor information	Hospital information card; language choice; cost shown before booking a direct test

New features added and where they fit in this document

Feature	What it does	Idea from	Where it fits
F1 Direct lab test	Patient (or front desk for a walk-in) books tests from an approved list without a doctor visit; abnormal result → advice to see a doctor	Team requirement; sites offer only fixed packages	Fig 1 (5b), Fig 3, Fig 5 (flows 3, 9, 22), Fig 6C, access matrix, risks, tests
F2 Hospital and doctor directory	Search by hospital / branch, department or doctor; shows hours, emergency number and free slots	Ibn Sina branch search; Square satellite centres	Fig 3 (hospital, department, practitioner_role), Fig 5 (D6)
F3 Test packages	Bundles of approved direct tests (e.g. basic health check)	Aurora, Evercare, Ibn Sina packages	lab_test list in D6
F4 Live queue token	Patient sees token number and how many patients are ahead	Not seen on the sites checked	Fig 5 (flows 6, 30), feature table
F5 One login	Booking, reports and prescriptions under one patient account	BIRDEM, Evercare, Square use separate portals	Fig 2, Section 7
F6 Bangla / English	Screens and SMS in both languages	Ibn Sina (Bangla notices), Mayo (language choice)	Feature table
F7 Emergency numbers on hospital card	Hospital emergency and ambulance numbers shown with every hospital	BIRDEM hotline list	Fig 3 (hospital.emergency_phone)

Future phases (not in this document’s scope): video consultation, online payment, medicine delivery (seen at SGH and Evercare).

How the direct lab test works

- **Approved list only.** Only tests marked “direct allowed” (for example blood sugar, CBC, lipid profile, urine routine, HbA1c; the final list is set by the hospital with its clinical advisor) can be booked without a doctor. Tests that need interpretation, imaging, special preparation or carry higher risk always need a doctor’s order.
- **No encounter needed.** The order is saved with type = direct and no encounter; the lab handles it like any other order.

- **Safe result release.** The patient sees the report only after the lab verifies it. Abnormal values are flagged and the patient gets a “please see a doctor” message with a booking link. A critical value makes the lab verifier phone the patient and alert the duty doctor.
- **One record.** The result is stored in the patient record, so the doctor sees it at the next visit.

6. Methodology

The system is built in short Agile sprints with a clinician advisor reviewing safety features: requirements → design → core platform → AI pipeline and offline validation → security and usability testing → pilot. Figure 1 (next page) shows the complete patient journey: **8 stages (plus the direct lab path 5b), 7 decision points and 4 care paths** (standard OPD, urgent OPD, Emergency Room, direct lab test).

How each feature works

Feature	Trigger	How it works	Output
Hospital & doctor search	Patient opens the app or site	Search by hospital / branch, department or doctor; slots come from the directory	Hospital card (address, hours, emergency number) + free slots
Registration	New user signs up	Collect demographics and consent; verify phone / email	Authenticated profile
Appointment booking	Patient picks doctor and slot	Check slot availability → create appointment	Confirmation + reminder to both
Direct lab test	Patient (or front desk) chooses “Book a lab test”	Pick a test or package from the approved list and a slot; lab collects sample, uploads and verifies report	Report in record + patient alert; “see a doctor” advice if abnormal
Queue token	Patient is checked in	Front desk checks in; system issues a token and shows position	Token + patients ahead (phone or screen)
Adaptive emergency queue	Emergency flag or critical vitals	Critical → Red Alert + ER routing; Urgent → moved ahead of standard patients	ER alert, or updated queue with priority
Vitals capture	Patient is called by the nurse	Nurse enters BP, pulse, temperature, SpO2, respiratory rate; each one is classified	Classified vitals + trend chart; critical value escalates
Doctor consultation	Patient is called	Doctor sees history, allergies, vitals trend, lab reports; records diagnosis (ICD-10) and notes	Encounter record
Doctor-ordered lab test	Doctor orders a test	Lab receives order, collects sample, uploads and verifies report	Report in record + automatic doctor notification
AI e-prescription	Doctor submits typed draft or photo	AI reads, matches medicine, checks safety, scores each item	Draft with pre-filled items, highlighted fields and flags (shown live)
Pharmacy dispensing	Prescription is signed	Pharmacist validates, checks stock, dispenses	Dispensing record + status returned
Follow-up & reminders	Prescription / follow-up date	Scheduler creates medicine, visit and report reminders (SMS, email, push; Bangla or English)	Reminder + adherence log

Module decomposition (10 modules, 4 branches)

Branch	Modules
Patient	M1 Registration & profile · M2 Hospital & doctor search, appointment, triage, Red Alert & queue · M8 Follow-up & notifications
Physician	M3 Consultation & vitals · M4 AI prescription engine
Laboratory & Pharmacy	M5 Lab coordination (doctor-ordered and direct tests) · M6 Pharmacy & dispensing
Platform	M7 Identity, roles & consent · M9 Audit & compliance · M10 Admin analytics & configuration (hospital directory, direct-test list, DHIS2 export)

6.2 End-to-end flowchart and emergency decision

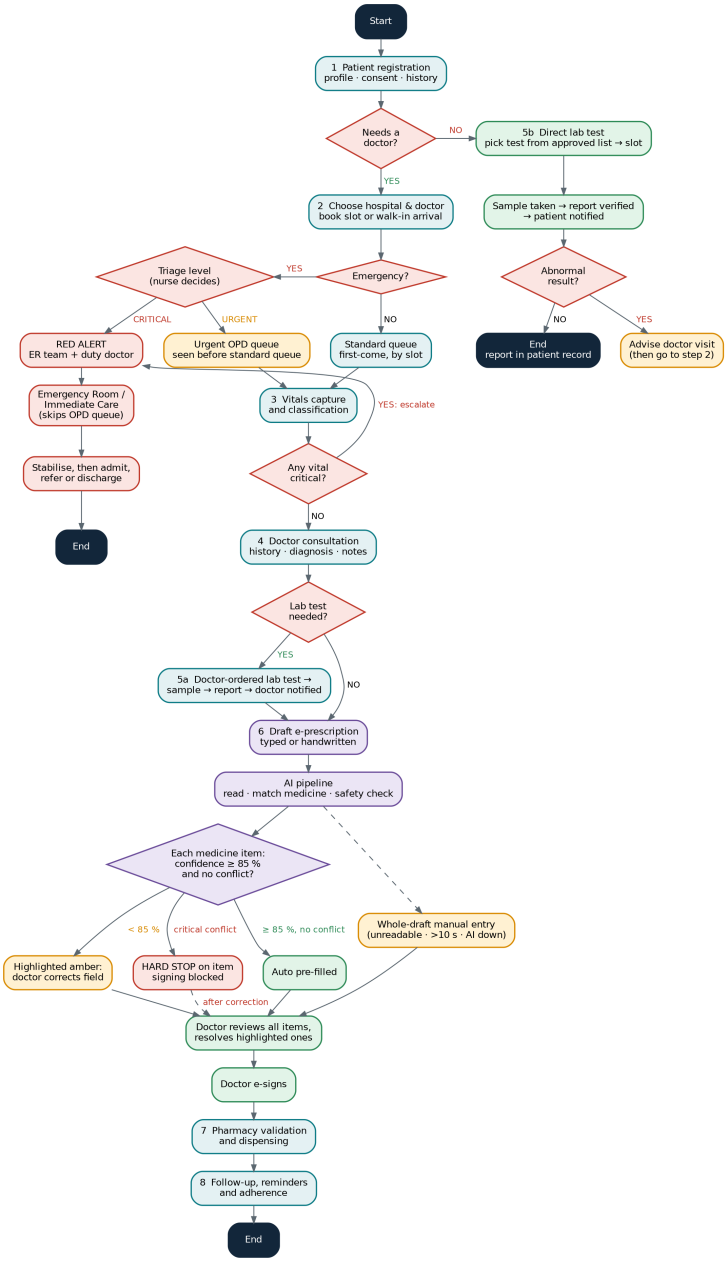


Figure 1. End-to-end flowchart with the direct lab path, emergency branching and the per-item 85 % confidence gate

How the emergency decision works

1. Emergency check. At booking or arrival, the front desk or nurse marks the case as emergency or not.

2. Not an emergency: standard queue (first-come, by slot), then vitals.

3. Emergency: the nurse chooses the triage level.

Critical (e.g. severe trauma): **Red Alert** notifies the ER team and duty doctor; the patient goes directly to the Emergency Room and never enters the OPD queue.

Urgent, not critical: stays in OPD, seen before standard patients.

4. Safety net. If any OPD patient's vitals are classified critical, the system escalates to Red Alert automatically.

5. After the ER branch, the case leaves this system's scope (admit, refer or discharge).

Who is seen first? (urgent queue)

The **nurse** decides the level for every emergency case. No formula is used:

Level	What happens
Critical	Red Alert → ER team and duty doctor; patient goes straight to the Emergency Room
Urgent	Stays in OPD, seen before standard patients. Among urgent patients, those with borderline vitals first, then by arrival time
Normal	Standard queue: first-come, by slot

Any critical vital sign automatically raises a Red Alert, even if the nurse did not mark the case as critical. The nurse can always re-order the queue; every change is logged.

Direct lab test path (5b)

A patient who does not need a doctor books a test from the approved list. After the lab verifies the report the patient is notified. If a value is abnormal the patient is advised to see a doctor and goes to step 2; otherwise the journey ends with the report in the patient record.

7. System Architecture

MediFlow AI is one main web application (a “modular monolith”) for all clinical workflows, plus a separate Python AI service that works in the background. One application is simpler to run and secure than many small services, and the clear modules (M1–M10) can be split later if needed. Patients, staff and lab use one login system, so booking, reports and prescriptions are not spread over separate portals.

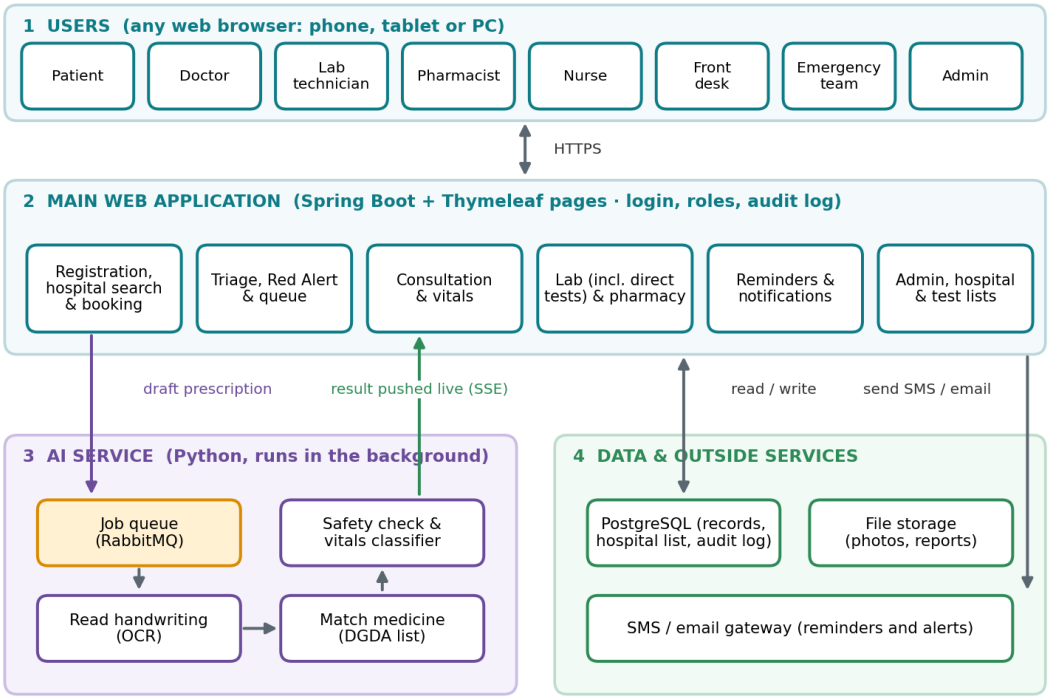


Figure 2. System architecture. AI results reach the doctor’s screen through live updates (SSE), so the screen never freezes while the AI works

Layer	Technology (medium-level, widely used)
Backend + pages	Java, Spring Boot, Thymeleaf (server-rendered pages), Bootstrap
Security	Spring Security: login, roles, MFA for doctors / pharmacists / admins
AI service	Python (FastAPI), ready-made OCR (e.g. PaddleOCR / Tesseract), simple medicine-name matching
Background jobs & live updates	RabbitMQ job queue; Server-Sent Events (SSE) to push results to the browser
Database & files	PostgreSQL (patient records, hospital and test directory); file storage for prescription photos and report PDFs
Notifications	SMS gateway, email (SMTP), push messages
Deployment	Docker, GitHub Actions, basic monitoring

Web or Android? Web-first: works on any phone or PC without installation, one code base for all roles, instant updates, and a smaller security surface. An Android app is a phase-2 option using the same back end.

8. Database Design

Green tables are new in this edition: the hospital directory (so a patient can choose where to go) and the lab test list (so direct tests are possible).

Table	Key columns	Relationship
hospital	id, name, type, address, phone, emergency_phone, opening_hours	1 hospital → many departments, doctor roles, appointments, encounters, lab orders
department	id, hospital_id, name	belongs to 1 hospital
practitioner	id, name, role, licence_no	1 practitioner → many roles (a doctor can work at several hospitals), many encounters
practitioner_role	id, practitioner_id, hospital_id, department_id, schedule	links doctor ↔ hospital ↔ department; schedule gives the bookable slots
patient	id, name, dob, sex, phone, national_id (encrypted)	1 patient → many appointments, encounters, lab orders, allergies
appointment	id, patient_id, doctor_id, hospital_id, slot, priority_level, emergency_level	1 appointment → 0..1 encounter
encounter	id, patient_id, doctor_id, hospital_id, appointment_id, started_at, diagnosis	1 encounter → **many** prescriptions, vitals, lab orders; 0..1 red alert
red_alert / vital_observation	red_alert: encounter_id, level, raised_by, time · vital: encounter_id, type, value, unit, class	belong to 1 encounter
lab_test	id, name, loinc_code, direct_allowed (yes / no)	1 test → many lab orders

Table	Key columns	Relationship
lab_order	id, patient_id, hospital_id, encounter_id (optional), test_id, order_type, status	doctor-ordered: has an encounter; direct test: no encounter (order_type = direct)
diagnostic_report	id, lab_order_id, file_ref, verified_by, abnormal_flag	1 order → 0..1 report
prescription	id, encounter_id, status, signed_at	**many per encounter**; 1 → many items, many dispenses
prescription_item	id, prescription_id, brand_id (optional), generic_id, dose, frequency, duration, confidence, review_status	brand optional, generic always set, so a generic-name prescription is accepted
drug_brand / drug_generic	brand: name, dgda_reg_no, strength, form, generic_id · generic: name, atc_code, rxnorm_code	many brands → 1 generic
allergy, dispense, audit_event	linked to patient / prescription; audit_event is append-only	—

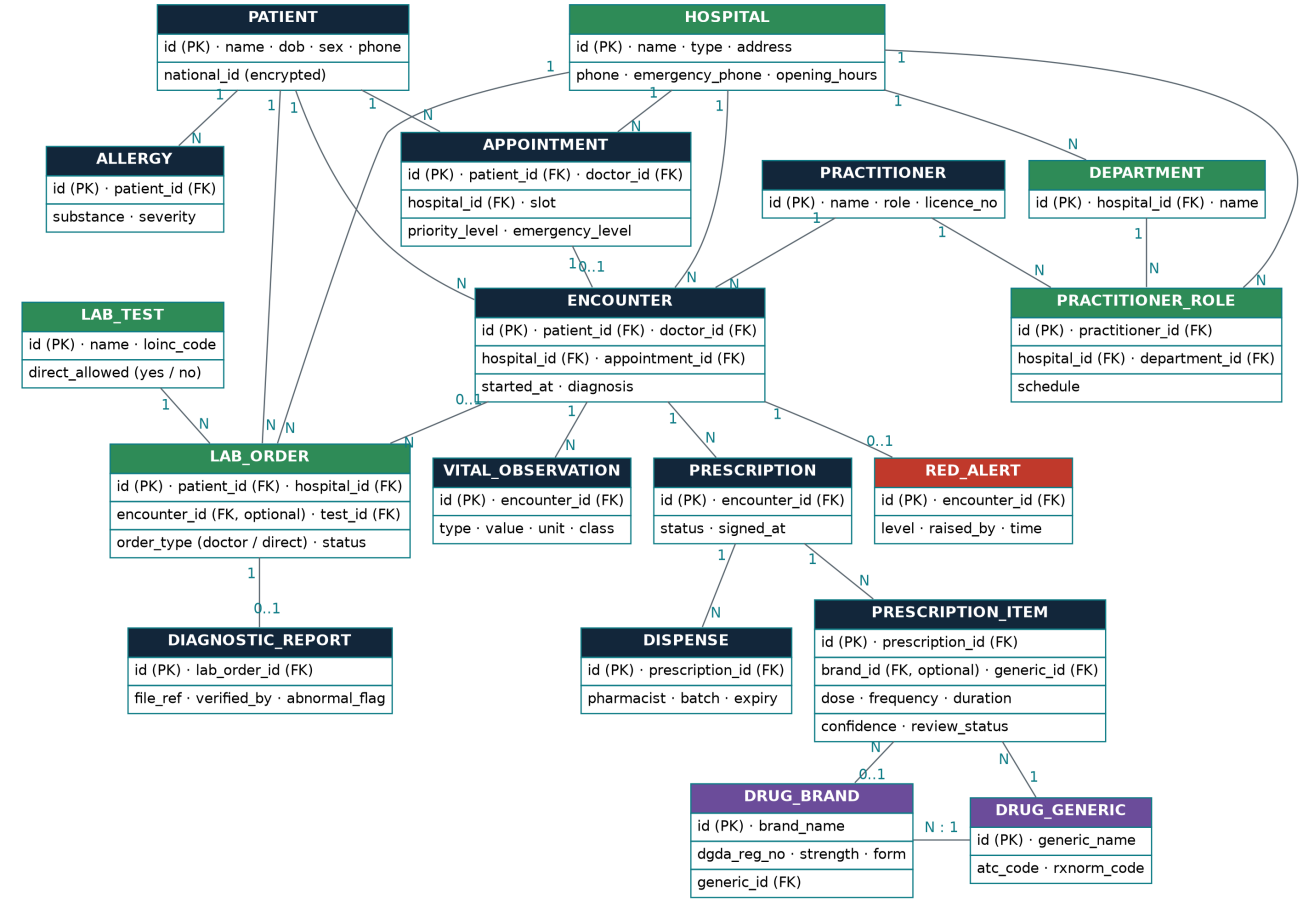


Figure 3. Entity-relationship diagram (green = hospital directory and lab tables). A direct lab test has no encounter; an encounter can have many prescriptions; an item links to a generic (always) and a brand (optional)

9. Data Flow and Behaviour

9.1 DFD Level 0 – context diagram

MediFlow AI is shown as one process exchanging data with eight external entities. **Nurse** (triage level, vitals) and **Front desk** (registration, booking, direct-test booking, check-in) are separate roles; the **Emergency team** receives the Red Alert and returns the handover status.

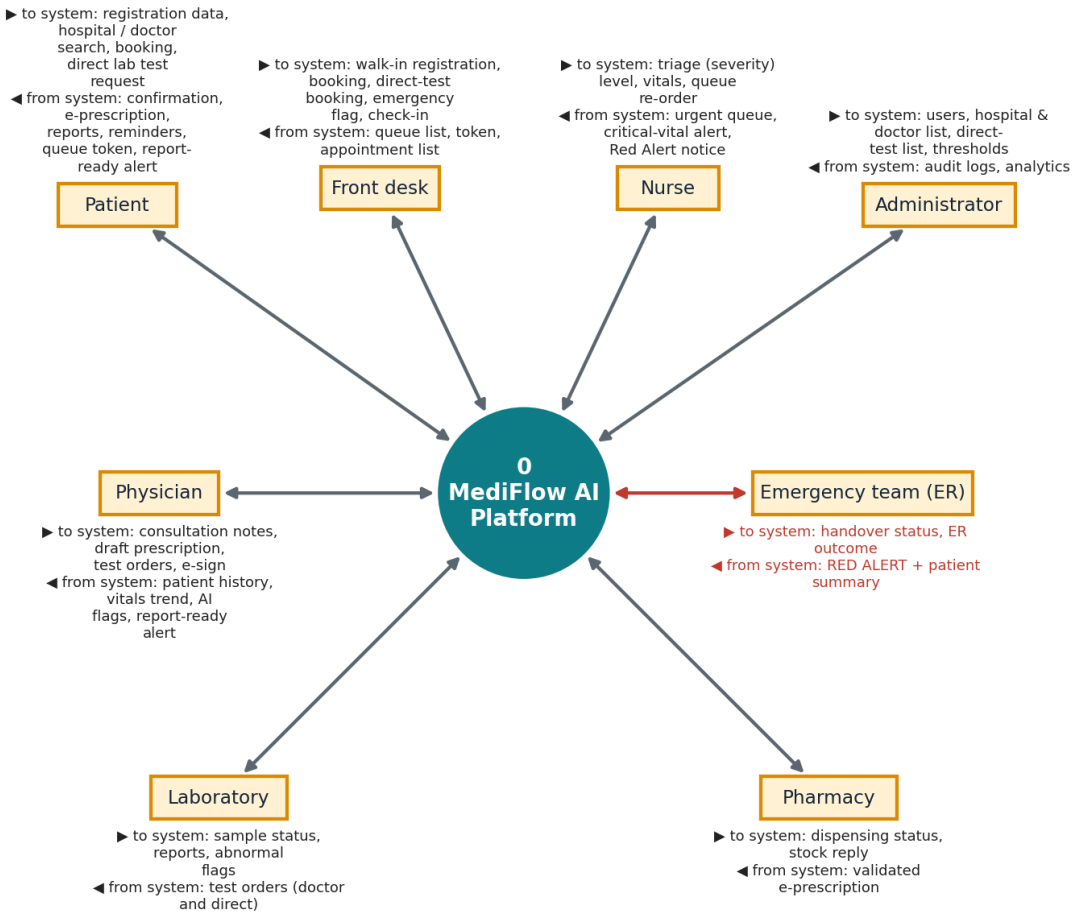


Figure 4. DFD Level 0 – context diagram

Data stores (used in Level 1)

Store	Content
D1 Patient records	Demographics, history, allergies, vitals, encounters (FHIR)
D2 Appointments, queue & alerts	Bookings, queue state, priority level, emergency level, Red Alert records
D3 Prescriptions	Drafts, signed e-prescriptions, AI flags, dispensing status
D4 Lab orders & reports	Doctor-ordered and direct test orders, samples, diagnostic reports
D5 Audit log	Append-only record of every access and change (all processes write)
D6 Hospital & test directory	Hospitals, departments, doctors and schedules, lab test list with the direct-allowed flag (edited by the administrator)

9.2 DFD Level 1 – processes and data stores

Level 1 has **eight processes** and six data stores. Process **8.0 Red Alert & ER Dispatch** receives critical triage (flow 8) and critical vitals (flow 12), reads the patient summary, alerts the ER team and nurse, and records the alert. New in this edition: the Front desk entity, store **D6 Hospital & test directory**, the direct-test booking flow (9) and the report-ready alert for direct tests (22).

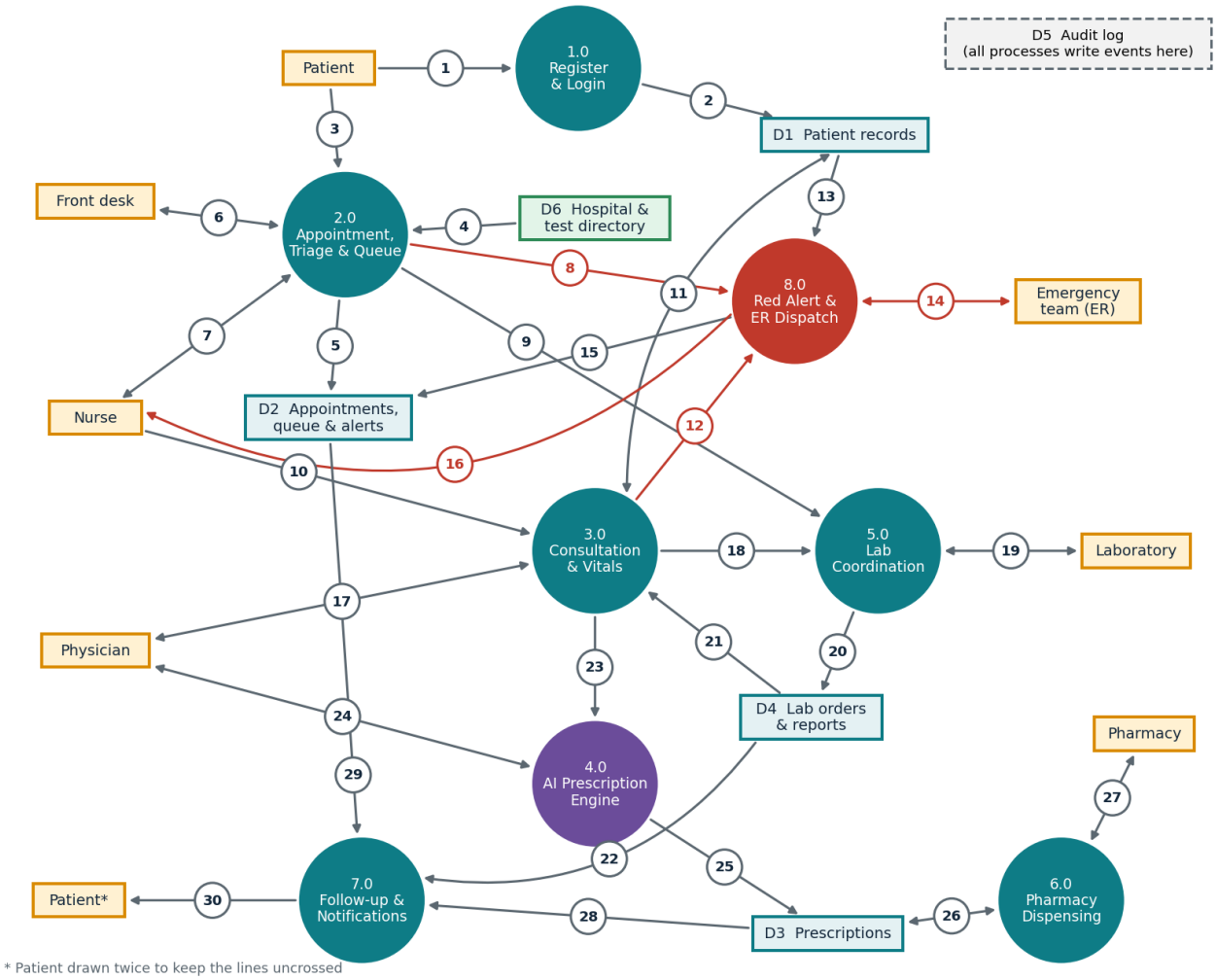


Figure 5. DFD Level 1 (numbers refer to the flow list below; red = emergency flows). Every process also writes events to D5 Audit log

1 Patient → 1.0: profile, consent	16 8.0 → Nurse: red alert notification
2 1.0 → D1: patient record	17 Physician ⇄ 3.0: notes, diagnosis / history, vitals trend
3 Patient → 2.0: hospital / doctor search, booking, direct lab test request	18 3.0 → 5.0: doctor test order
4 D6 → 2.0: hospitals, doctors, slots, direct-test list	19 5.0 ⇄ Lab: order / sample status, report
5 2.0 → D2: booking, queue	20 5.0 → D4: orders, reports
6 Front desk ⇄ 2.0: walk-in, booking, emergency flag / queue list	21 D4 → 3.0: report-ready alert (doctor-ordered)
7 Nurse ⇄ 2.0: severity level / urgent queue list	22 D4 → 7.0: report-ready alert (direct tests)
8 2.0 → 8.0: CRITICAL triage	23 3.0 → 4.0: draft Rx + history, allergies
9 2.0 → 5.0: direct test booking	24 4.0 ⇄ Physician: AI flags / corrections, e-sign
10 Nurse → 3.0: vitals	25 4.0 → D3: signed Rx
11 3.0 ⇄ D1: vitals, encounter / history, allergies	26 D3 ⇄ 6.0: signed Rx / dispense status
12 3.0 → 8.0: CRITICAL vitals	27 6.0 ⇄ Pharmacy: validated e-Rx / dispensing status
13 D1 → 8.0: patient summary	28 D3 → 7.0: follow-up dates
14 8.0 ⇄ ER: RED ALERT + summary / handover status	29 D2 → 7.0: appointments
15 8.0 → D2: alert record	30 7.0 → Patient: reminders, report ready, “see a doctor” advice

9.3 Sequence diagrams

Figure 6A: doctor-ordered lab workflow. Figure 6B: AI-assisted prescription. Figure 6C: direct lab test.

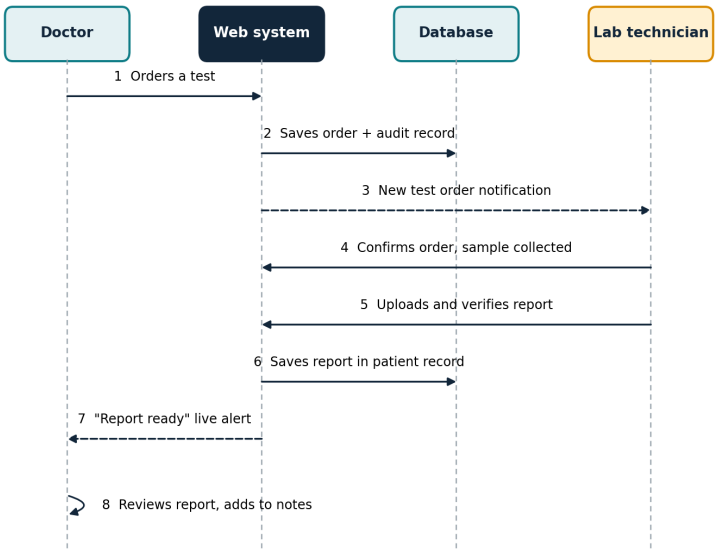


Figure 6A. Doctor-ordered lab workflow: order → sample → report → doctor notified

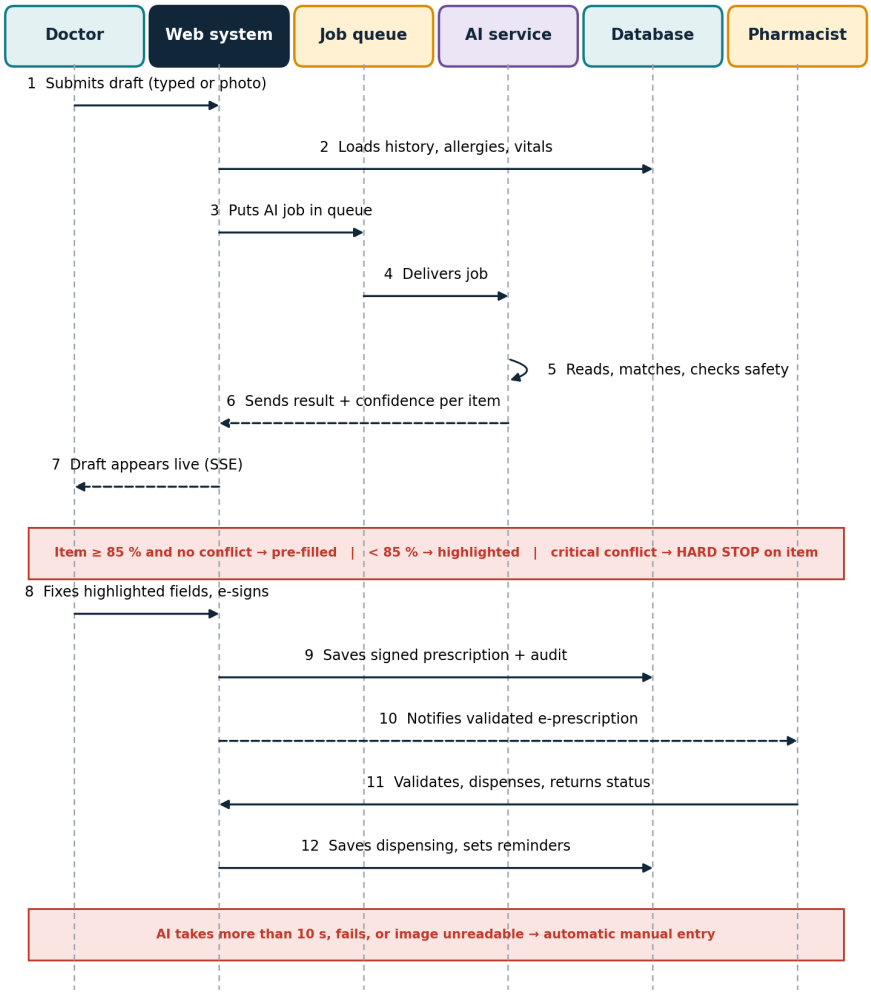


Figure 6B. AI-assisted prescription: the web system sends the job through the queue (as in Figure 2); the result returns and is shown live

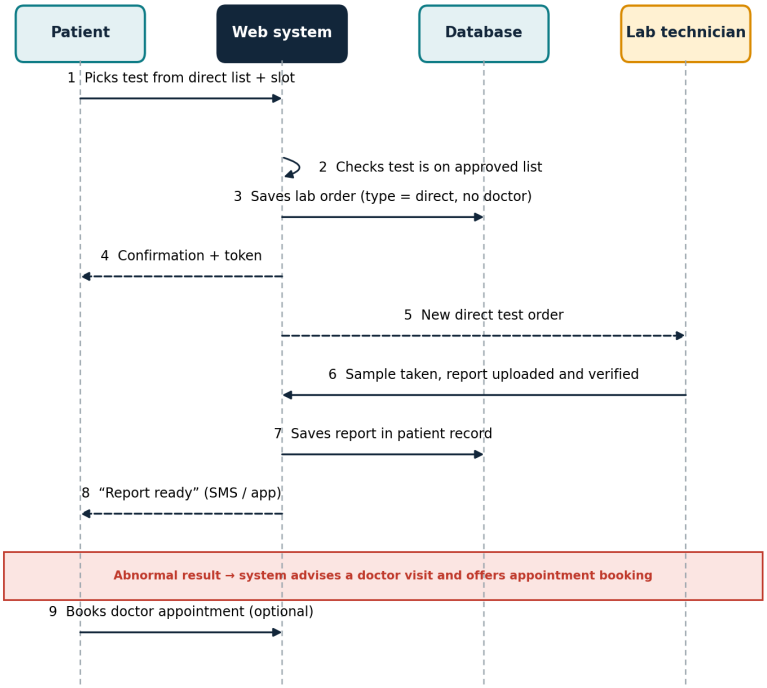


Figure 6C. Direct lab test: no doctor order; abnormal result leads to advice to see a doctor

10. Data Standards and Bangladesh Context

HL7 FHIR R4 is the common data format, so records can move between facilities and national systems. Main mappings: Patient → *Patient*; hospital → *Organization / Location*; doctor at a hospital → *PractitionerRole*; consultation → *Encounter*; Red Alert → *Encounter (emergency)* + *Flag*; vitals → *Observation* (LOINC, UCUM); diagnosis → *Condition* (ICD-10); prescription → *MedicationRequest*; dispensing → *MedicationDispense*; lab (doctor-ordered or direct) → *ServiceRequest / DiagnosticReport* (a direct test has the patient as requester); audit → *AuditEvent*.

Brand-first medicine mapping (DGDA)

Local doctors prescribe brand names, so every medicine is resolved as: **brand as written → DGDA-registered product → generic name → WHO ATC / RxNorm code**. Safety checks run on the generic level, so two brands of paracetamol are flagged as duplicate therapy. A brand that cannot be mapped is treated as **unknown, never safe**.

Brand written	Generic	Example strength / form	WHO ATC
Napa	Paracetamol	500 mg tablet	N02BE01
Seclo	Omeprazole	20 mg capsule	A02BC01
Ciprocin	Ciprofloxacin	500 mg tablet	J01MA02

Illustrative rows only; real strengths and registration numbers must come from the current DGDA drug register.

National alignment. DGHS: follow DGHS reporting formats and facility identifiers (to be confirmed before the pilot). **DHIS2:** export only *anonymised, aggregated* indicators (visits, lab turnaround, prescriptions issued, emergency cases) from module M10. Patient-level exchange stays on FHIR R4 with patient consent. Live integration depends on DGHS approval.

11. AI Engineering

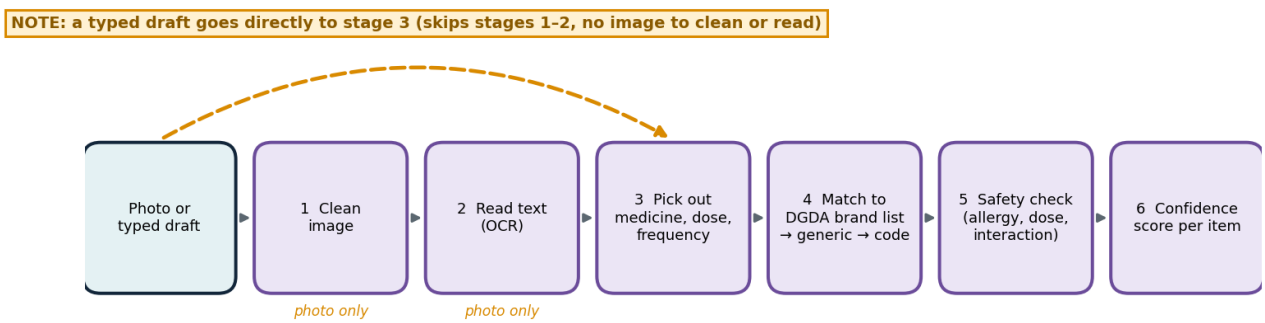


Figure 7. AI pipeline. Photos go through all stages; a typed draft goes directly to stage 3

Handwriting is read with a ready-made OCR engine fine-tuned on consented, anonymised prescription images; medicine names are matched to the DGDA brand list; each field gets a confidence score between 0 and 1. The AI never writes to the record directly: its output is

a draft shown next to the original image. Jobs run in the background (the screen stays responsive); if SSE drops the screen polls the job status, and after 10 s without a result it switches to manual entry.

The 85 % rule (applied to each medicine item)

> **Item confidence = the lowest of (drug, dose, frequency) confidence.** If ≥ 0.85 and no conflict → pre-filled. If < 0.85 → highlighted. Critical conflict → hard stop.

Item result	What the doctor sees	Can it be signed?
≥ 85 % and no conflict	Pre-filled automatically (doctor still reviews)	Yes, after review
< 85 %	Field highlighted amber; AI guess shown beside the original image; doctor confirms or corrects	Only after the doctor resolves it
Critical conflict (allergy, severe interaction, lethal dose) at any confidence	Hard stop on that item	No, until corrected
Medicine not found in DGDA list	Treated as < 85 %: highlighted and left blank	Only after the doctor enters it

The **whole draft** goes to manual entry only when the image is unreadable, the AI takes more than 10 s, or the service is down. Nothing is ever auto-signed or auto-dispensed.

Contradiction checks (every draft)

Check	Example	Severity → action
Allergy conflict	Penicillin for a penicillin-allergic patient	Critical → hard stop on item
Severe drug–drug interaction	Two interacting drugs on the active list	Critical → hard stop; change or override with reason
Dose out of range	Paediatric dose above age/weight limit	Critical → hard stop
Duplicate therapy	Same generic under two brands	Warning → confirm
Condition / pregnancy / vitals caution	NSAID with renal impairment; antihypertensive increase with low BP	Warning → confirm
Unknown item	Brand not in DGDA list	Unknown → treated as < 85 %, highlighted

Vital-sign classification: one design for every vital

Vital signs arrive in different shapes: heart rate is a number (72) while blood pressure is text (“120/80”). Writing a separate variable and check for each vital would repeat the same code again and again. MediFlow AI therefore treats every vital as one type, **VitalSign**, and uses **polymorphism** (Figure 8):

- Each kind of vital knows how to read its own input (*parse*, from a number or from text) and how to classify itself (*classify*): **NumericVital** for heart rate, respiratory rate, temperature and SpO2; **BloodPressureVital** for blood pressure.
- The rest of the system only calls *v.classify()*. It never needs to know whether the value is a number or text, so no new variable is declared for each vital.
- All limits live in one rule table (*VitalRule*). Adding a new vital, for example blood glucose, means adding one row, not new code.

> **Level(v) = CRITICAL** if v is in the critical range; **BORDERLINE** if v is in the borderline range; otherwise **NORMAL**.
> **Patient level** = the highest level among all vitals (NORMAL < BORDERLINE < CRITICAL).

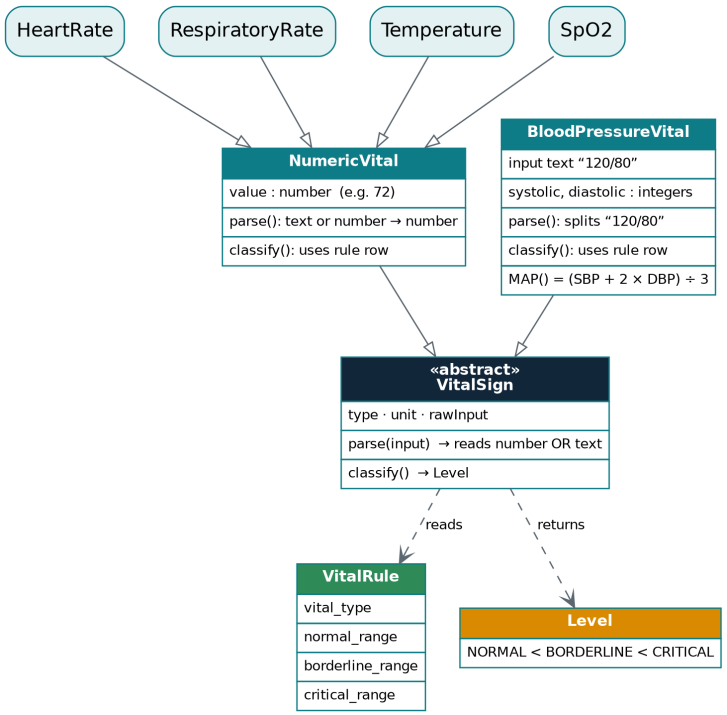


Figure 8. Vital-sign design: one base type, subclasses that parse their own input, and one shared rule table

Formula	Meaning	Use in classification (defaults; clinical advisor confirms)
MAP = (SBP + 2 × DBP) ÷ 3	Mean arterial pressure from blood pressure	MAP < 65 mmHg → CRITICAL
Pulse pressure = SBP − DBP	Gap between systolic and diastolic pressure	Shown to the doctor (information only)
Shock index = HR ÷ SBP	Heart rate relative to systolic pressure	≥ 0.9 → BORDERLINE; ≥ 1.0 → CRITICAL

Rule table (adult default; configurable, clinician-approved)

Vital sign	Normal	Borderline	Critical
Blood pressure (mmHg)	< 120 / < 80	120–179 / 80–119	≥ 180 / ≥ 120, or systolic < 90
Heart rate (/min)	60–100	50–59 or 101–129	< 40 or ≥ 130
Respiratory rate (/min)	12–20	9–11 or 21–24	≤ 8 or ≥ 25
Temperature (°C)	36.1–37.2	37.3–39.4 or 35.1–36.0	≥ 39.5 or ≤ 35.0
SpO2 (%)	≥ 95	91–94	≤ 90

A critical reading alerts the nurse and doctor and, in OPD, escalates the patient to Red Alert. Every classified reading is stored, so the doctor sees a trend. Paediatric and pregnancy tables are separate.

Evaluation (design targets to be confirmed in testing)

Component	Metric	Target
Handwriting OCR	Character error rate	≤ 2 % printed; ≤ 10 % handwritten
Medicine matching (DGDA)	Top-1 / Top-3 accuracy	≥ 95 % / ≥ 98 %
Auto-filled fields	Precision per field (drug, dose, frequency)	≥ 99 % (to set with clinical advisor)
Contradiction engine	Recall on critical conflicts / overall precision	≥ 98 % / ≥ 80 %
Vitals classifier	Critical-case sensitivity	≥ 99 %
Confidence calibration	So that “85 %” really means about 85 %	Calibration error ≤ 0.05
Speed	AI end-to-end (95th percentile)	≤ 5 s

Models are tested offline on handwriting from unseen writers, then in **shadow mode** (AI runs, results hidden) in the pilot before any clinical use. The 85 % threshold can only be raised, never lowered, and only with clinical approval.

12. Security and Access Control

- **Login:** hashed passwords, MFA for doctors, pharmacists and admins, session time-out. **Access:** role-based, deny by default; doctors see only assigned patients.
- **Protection:** HTTPS everywhere, encrypted database and files, encrypted national ID and contact fields, standard web protections (CSRF, input validation).
- **Audit:** every read and write is logged in an append-only audit log; patients can see who accessed their record.
- **Emergency access:** the ER team gets access to a Red-Alert patient; other emergency access uses “break-glass” with a mandatory reason and post-review.
- **AI privacy:** images and text are never sent to outside AI services; models run in the private network.

Access matrix

Resource	Patient	Doctor	Lab tech	Pharmacist	Nurse	Front desk	Admin
Demographics	R U[1]	R	R*	R*	R*	R C U	—
History & allergies	R C[2]	R C U	—	R†	—	—	—
Vital signs	R C[2]	R C U	—	—	R C	—	—
Appointments & queue	R C	R U	—	—	R U	R C U	R‡
Hospital & test directory	R	R	R	R	R	R	C U[5]
Consultation notes	R[3]	R C U	—	—	—	—	—
Draft prescription	—	R C U	—	—	—	—	—
Signed e-prescription	R	R S	—	R D	—	—	—
Lab orders	R C[6]	R C	R U	—	R	C[6]	—
Lab reports	R	R	R C U V	—	—	—	—
Dispensing record	R	R	—	R C U	—	—	—
AI flags & override log	—	R O	—	R	—	—	R‡
Audit log	R[4]	—	—	—	—	—	R
Settings & users	—	—	—	—	—	—	C U[5]

R read · C create · U update · S e-sign · D dispense · V verify · O override with reason · — no access. [1] own contact details only (name / date-of-birth changes need a verified request) · [2] self-reported, “unverified” until doctor confirms · [3] summary only · [4] own access history · [5] admin edits the hospital directory and direct-test list; clinical rules / threshold changes need dual approval · [6] direct tests from the approved list only (front desk books on the patient’s behalf) · * minimum data · † allergies and active medicines · ‡ aggregated data. **ER team:** doctor-level access limited to the Red-Alert patient they are assigned.

13. Risk Management and Clinical Fallback

Risk	Protection	Fallback
AI misreads drug or dose	85 % per-item gate, highlighted fields, original image shown beside result, doctor review, pharmacist second check	Doctor corrects or enters manually; incident logged
Allergy / interaction missed	Rule-based checks on generic level; unknown = unverified	Hard stop on item; manual review
AI service down or slow	Job queue with retries, health checks, 10 s time-out	Automatic switch to manual prescription
Live updates (SSE) lost	Automatic reconnect, status polling	Polling; after 10 s manual entry
Emergency mis-prioritised	Nurse sets the level, critical vitals auto-escalate, nurse override, all changes logged	Manual queue control; ER called directly
Direct test misused or abnormal result ignored	Approved list only; report released after lab verification; abnormal flag + “see a doctor” advice; critical value phoned	Lab verifier calls the patient; duty doctor alerted
Wrong patient identity	Two-identifier match (ID + date of birth); QR token at pharmacy and lab	Block until identity confirmed
Hospital or doctor list outdated	Admin-managed directory with last-updated date; schedule change alerts booked patients	Front desk confirms by phone
Unmapped / outdated brand	Scheduled DGDA refresh; unmapped brand = unknown	Doctor selects medicine; pharmacist verifies
Data breach / misuse	Encryption, roles, MFA, audit alerts	Incident response, access revoked
Network loss at clinic	Short-term local cache (planned)	Paper downtime procedure, back-entry later

Six barriers against a wrong medicine: (1) confidence gate, (2) DGDA dictionary check, (3) contradiction engine, (4) doctor review beside the original photo, (5) independent pharmacist validation, (6) patient copy with brand and generic names. **If an error is found after issue,** the prescription is revoked, doctor, pharmacy and patient are notified, the case is reviewed within 24 hours, and the model can be rolled back or the threshold raised.

14. Testing, Expected Outcomes and Limitations

Test level	What is tested
Unit / integration	Rule engine, FHIR mapping, brand-to-generic mapping, vital-sign classes (number and text input)
Security	Login, every cell of the access matrix, common web attacks
AI offline + shadow mode	Metrics above, including per-field precision; clinician reviews errors
Emergency and lab flows	Critical vs urgent routing, auto-escalation on critical vitals; doctor-ordered and direct test: order → sample → report → notification
Direct test safety	Only approved tests bookable; abnormal and critical results trigger advice / phone call
Hospital directory	Search by hospital, branch, department, doctor; slot accuracy; emergency numbers shown
Live updates and speed	SSE delivery, reconnect, polling, 10 s manual switch, many users at once
Usability and failure drills	Doctor / nurse / front desk / patient task success; AI outage, SSE loss, notification failure

Expected outcome	How it is measured (KPI)
Better coordination and continuity	Appointment-to-consultation time; share of visits with a complete linked record
Fewer prescription errors	Prescriptions needing pharmacist clarification; errors caught before dispensing
Useful AI assistance	Share of AI fields accepted unchanged; correction rate on highlighted fields; time saved
Efficient lab workflow	Order-to-report time (doctor-ordered and direct); report-to-doctor-review time
Easy access for patients	Share of direct tests booked without a doctor visit; share of abnormal direct results followed by a doctor visit; booking completion time
Safe emergency handling	Arrival-to-ER time for critical cases; arrival-to-triage time for urgent cases
Better adherence	Reminder acknowledgement and follow-up attendance rate

Limitations and future work

- AI accuracy depends on handwriting data quality and variety; pilot validation is required before clinical use.
- The direct-test list and the abnormal-value limits need clinical approval; interaction rules need a licensed or curated source; the DGDA register must be obtained and refreshed.
- The hospital directory needs a named owner at each hospital to stay up to date. DGHS and DHIS2 integration depends on approvals; offline operation is only partly covered.
- **Future:** Android app, video consultation, online payment, medicine delivery, wider clinical decision support, national system integration, population-health analytics.

Conclusion

MediFlow AI joins patients, doctors, laboratories and pharmacies in one platform. A hospital and doctor directory, direct lab tests for simple needs, emergency-aware routing, AI-assisted e-prescriptions with safety checks and smart follow-up improve coordination and patient safety, while the per-item 85 % gate with doctor review and pharmacist validation keeps the AI a safe assistant, not an unchecked decision-maker.

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